11 TRANSPORT IN PLANTS

Chapter 10 described how plants make food by photosynthesis. Photosynthesis requires water from the roots to be transported to the leaves, and sugars and other products of photosynthesis to be transported away from the leaves to the rest of the plant. This chapter explains how these materials are moved through the plant.

LEARNING OBJECTIVES

- Understand how water can move in or out of plant cells by osmosis
- Investigate osmosis in living and non-living systems (plant cells and Visking tubing)

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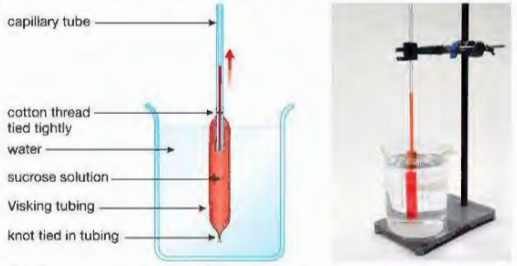
- Understand how water is absorbed by root hair cells
- Understand that transpiration is the evaporation of water from the surface of a plant
- Describe the role of the xylem in the transport of water and mineral ions from the roots to other parts of a plant
- Describe the role of phloem in the transport of sucrose and amino acids from the leaves to other parts of a plant
- Understand the origin of carbon dioxide and oxygen as waste products of metabolism and their loss from the stomata of a leaf

BIOLOGY ONLY

- Understand how the rate of transpiration is affected by changes in temperature, wind speed, humidity and light intensity
- Investigate the role of environmental factors in determining the rate of transpiration from a leafy shoot.

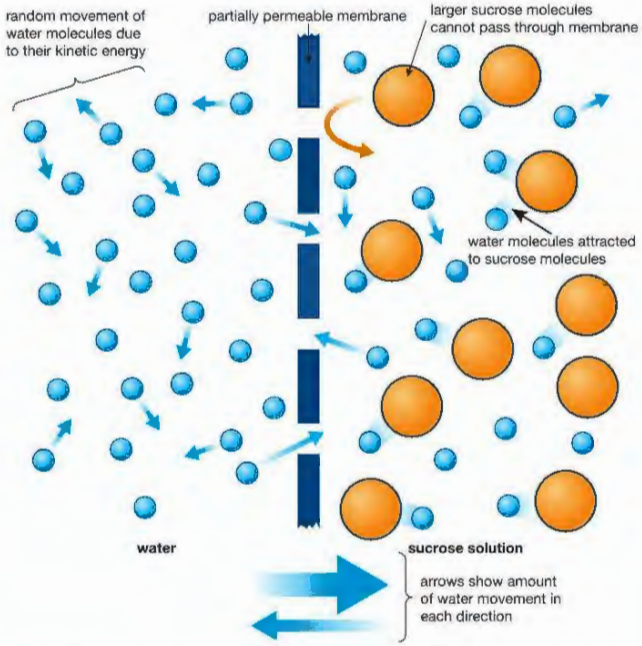
OSMOSIS

Osmosis is the name of a process by which water moves into and out of cells. In order to understand how water moves through a plant, you need to understand the mechanism of osmosis. Osmosis happens when a material called a **partially permeable membrane** separates two solutions. One artificial partially permeable membrane is called Visking tubing. This is used in kidney dialysis machines. Visking tubing has microscopic holes in it, which let small molecules like water pass through (it is *permeable* to them) but is not permeable to some larger molecules, such as the sugar sucrose. This is why it is called 'partially' permeable. You can show the effects of osmosis by filling a Visking tubing 'sausage' with concentrated sucrose solution, attaching it to a capillary tube and placing the Visking tubing in a beaker of water (Figure 11.1).



▲ Figure 11.1 Water enters the Visking tubing 'sausage' by osmosis. This causes the level of liquid in the capillary tube to rise. In the photograph, the contents of the Visking tubing have had a red dye added to make it easier to see the movement of the liquid.

The level in the capillary tube rises as water moves from the beaker to the inside of the Visking tubing. This movement is due to osmosis. You can understand what's happening if you imagine a highly magnified view of the Visking tubing separating the two liquids (Figure 11.2).



▲ Figure 11.2 In this model of osmosis, more water molecules diffuse from left to right than from right to left.

KEY POINT

Diffusion is the net movement of particles (molecules or ions) due to their kinetic energy, down a concentration gradient (see Chapter 1).

KEY POINT

It is important to realise that neither of the two solutions has to be pure water. As long as there is a difference in their concentrations (and their water potentials), and they are separated by a partially permeable membrane, osmosis can still take place.

The sucrose molecules are too big to pass through the holes in the partially permeable membrane. The water molecules can pass through the membrane in either direction, but those on the right are attracted to the sugar molecules. This slows them down and means that they are less free to move – they have less kinetic energy. As a result of this, more water molecules diffuse from left to right than from right to left. In other words, there is a greater diffusion of water molecules from the more dilute solution (in this case pure water) to the more concentrated solution.

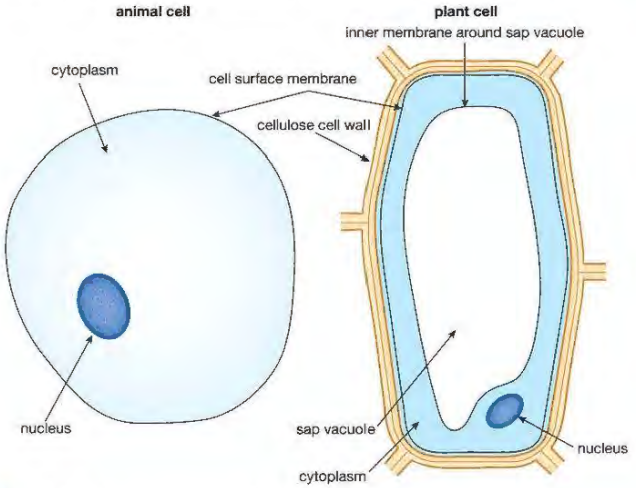
How 'free' the water molecules are to move is called the **water potential**. The molecules in pure water can move most freely, so pure water has the highest water potential. The more concentrated a solution is, the lower is its water potential. In the model in Figure 11.2, water moves from a high to a low water potential. This is a law which applies whenever water moves by osmosis. We can bring these ideas together in a definition of osmosis.

KEY POINT

Osmosis is the net diffusion of water across a partially permeable membrane, from a solution with a higher water potential to one with lower water potential.

OSMOSIS IN PLANT CELLS

So far we have only been dealing with osmosis through Visking tubing. However, there are partially permeable membranes in cells too. The cell surface membranes of both animal and plant cells are partially permeable, and so is the inner membrane around the plant cell's sap vacuole (Figure 11.3).



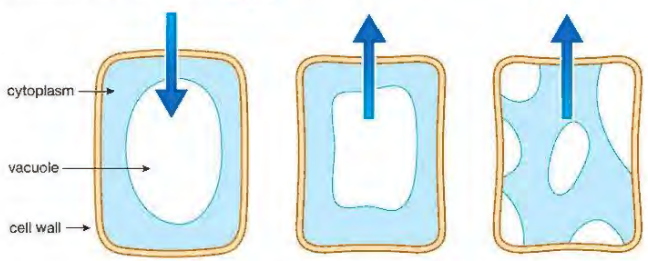
▲ Figure 11.3 Membranes in animal and plant cells

KEY POINT

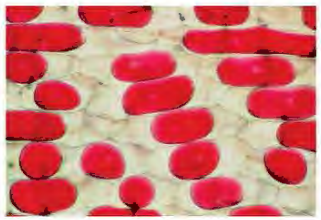
The cell *wall* has large holes in it, making it fully permeable to water and solutes. Only the cell *membranes* are partially permeable barriers that allow osmosis to take place.

Around the plant cell is the tough cellulose cell wall. This outer structure keeps the shape of the cell, and can resist changes in pressure inside the cell. This is very important, and critical in explaining the way that plants are supported. The cell contents, including the sap vacuole, contain many dissolved solutes, such as sugars and ions.

If a plant cell is put into pure water or a dilute solution, the contents of the cell have a lower water potential than the external solution, so the cell will absorb water by osmosis (Figure 11.4). The cell then swells up and the cytoplasm pushes against the cell wall. A plant cell that has developed an internal pressure like this is called **turgid**.



▲ Figure 11.4 The effects of osmosis on plant cells

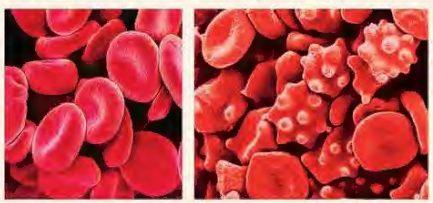


▲ Figure 11.5 Cells of onion epidermis, showing plasmolysis. The cell membranes and cytoplasm can be seen to have pulled away from the cell wall.

On the other hand, if the cell is placed in a concentrated sucrose solution that has a lower water potential than the cell contents, it will *lose* water by osmosis. The cell decreases in volume and the cytoplasm no longer pushes against the cell wall. In this state, the cell is called **flaccid**. Eventually the cell contents shrink so much that the membrane and cytoplasm split away from the cell wall and gaps appear between the wall and the membrane. A cell like this is called **plasmolysed**. You can see plasmolysis happening in the plant cells shown in Figure 11.5. The space between the cell wall and the cell surface membrane will now be filled with the sucrose solution.

KEY POINT

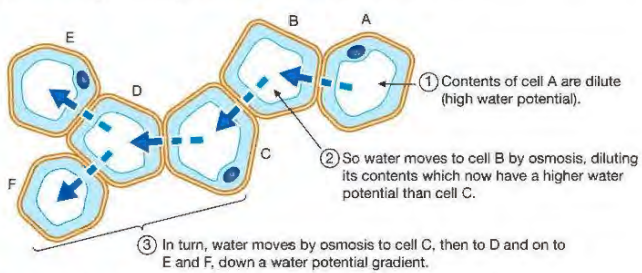
This chapter only covers plant cells. Osmosis also happens in animal cells, but there is much less water movement. This is because animal cells do not have a strong cell wall around them, and can't resist the changes in internal pressure resulting from large movements of water. For example, if red blood cells are put into water, they will swell up and burst. If the same cells are put into a concentrated salt solution, they lose water by osmosis and shrink, producing cells with crinkly edges (Figure 11.6).



◀ Figure 11.6 Compare the red blood cells on the right, which were placed in a 3% salt solution, with the normal cells on the left. Blood plasma has a concentration equal to a 0.85% salt solution.

Turgor (the state a plant is in when its cells are turgid) is very important to plants. The pressure inside the cells pushes neighbouring cells against each other, like a box full of inflated balloons. This supports the non-woody parts of the plant, such as young stems and leaves, and holds stems upright, so the leaves can carry out photosynthesis properly. Turgor is also important in the functioning of stomata. If a plant loses too much water from its cells so that they become flaccid, this makes the plant **wilt**. You can see wilting in a pot plant which has been left for too long without water. The leaves droop and collapse. In fact this is a protective action. It cuts down water loss by reducing the exposed surface area of the leaves and closing the stomata.

Inside the plant, water moves from cell to cell by osmosis. If a cell has a higher water potential than the cell next to it, water will move from the first cell to the second. In turn, this will dilute the contents of the second cell, so that it has a higher water potential than the next cell. In this way, water can move across a plant tissue, down a gradient of water potential (Figure 11.7).



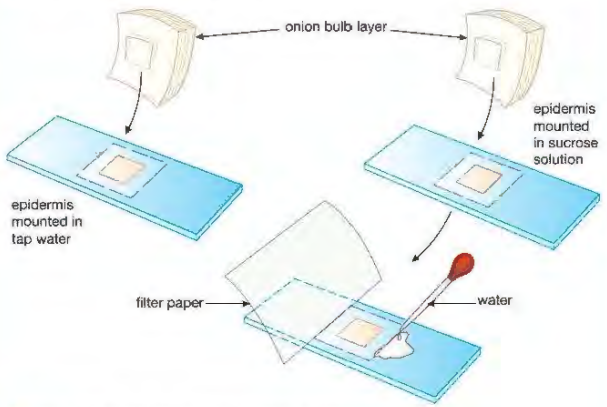
▲ Figure 11.7 Water moves from cell to cell down a water potential gradient.

Safety Note: Wash hands after handling and preparing the onion. Microscopes using a mirror for daylight illumination must never be placed where direct sunlight might strike the mirror otherwise permanent retinal damage may occur.

ACTIVITY 1

▼ PRACTICAL: INVESTIGATING THE EFFECTS OF OSMOSIS IN ONION EPIDERMIS CELLS

A drop of concentrated (1 mol per dm^3) sucrose solution is placed on one microscope slide, and a drop of distilled water on a second slide. Two small squares of inner epidermis are removed from the outer fleshy layers of an onion (Figure 11.8). One square is transferred to the sucrose solution, and the second to the water. This is done as quickly as possible, so that the cells do not dry out.



▲ Figure 11.8 Investigating osmosis in onion epidermis cells

A drop of the correct solution is added to the top of each specimen, followed by a cover slip. Any excess liquid is blotted (cleaned) up with filter paper.

Each slide is examined through the microscope for several minutes. The specimen in water will show turgid cells, similar to the first cell in Figure 11.4. The cells in sucrose solution will gradually plasmolyse, as shown in the last diagram of Figure 11.4, and Figure 11.5.

It is interesting to replace the sugar solution on the first slide with distilled water, to see the effect on the plasmolysed cells. This can be done quite easily without removing the cover slip. Some water is placed on one side of the cover slip and drawn across the slide using filter paper, as shown in Figure 11.8. After a while all the sucrose solution is replaced by water. If the cells are now observed for a few minutes, they will gradually recover from their plasmolysed condition until they are fully turgid again.

It may be possible for you to carry out these procedures. If you can, observe and draw one or two turgid cells, and one or two that are plasmolysed.

Can you explain all these observations in terms of osmosis?

If you use a red onion, the contents of the cells are coloured and easier to see.

Safety Note: Take great care when cutting the potato into 'chips' and wash hands afterwards.

ACTIVITY 2

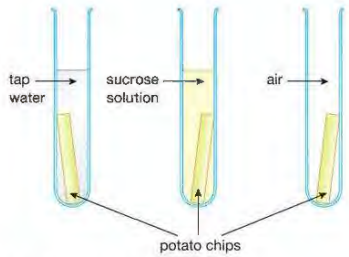
▼ PRACTICAL: INVESTIGATING THE EFFECTS OF OSMOSIS ON POTATO TUBER TISSUE

A potato tuber is a plant storage organ. It is a convenient tissue to use to investigate the effects of osmosis on the mass of the tissue.

A boiling tube is half filled with distilled water and a second with concentrated (1 mol per dm^3) sucrose solution. A third tube is left empty.

A potato is cut into chips 5 cm × 1 cm × 1 cm, making these measurements as accurate as possible so that the three chips are the same size. No skin is left on the potato tissue.

Each chip is gently blotted to remove excess moisture, and the mass of each is found by weighing on a balance. One chip is placed in each of the three boiling tubes (Figure 11.9).



▲ Figure 11.9 Investigating osmosis in potato tissue

After 30 minutes, the chips are removed from the tubes and gently blotted to remove excess liquid, then re-weighed. Each is felt in turn, to compare how flexible or stiff they are.

The change in mass (+ or –) of each chip is calculated, and the percentage change found from the equation:

$$\%\ \text{change} = \frac{\text{change in mass}}{\text{starting mass}} \times 100$$

The table below shows a set of results from this experiment.

Tube contents	Starting mass / g	Final mass / g	Change in mass / g	Change in mass / %	Condition (flexible / stiff)
water	6.25	6.76	+0.51	+8.16	stiff
sucrose solution	7.10	6.31	–0.79	–11.1	flexible
air	6.66	6.57	–0.09	–1.35	stiff

What conclusions can you make from these results? Explain them in terms of osmosis.

How large are the percentage changes in mass of the chips in the two liquids, compared with the chip in air?

Can you explain the final 'condition' of the chips, using terms such as 'flaccid' and 'turgid'?

Can you think of any criticisms of this experiment? For example, does using one chip per tube yield reliable evidence?

You may be able to perform an experiment similar to the one above. You don't have to use potato; other tissues such as carrot or sweet potato should work. An alternative to weighing is to measure the length of the chips at the end of the experiment, and compare these measurements with the starting lengths. However, weighing is much more accurate and gives a bigger percentage change.

You could try extending the experiment into a more complete investigation. For example you could make predictions about what would happen to chips placed in solutions that are intermediate in concentration between water and molar sucrose solution.

Plan a method to test your hypothesis. You should include a list of the apparatus and materials to be used, a description of the procedure, and a statement of the expected results.



▲ Figure 11.10 Root hairs on the growing root of a radish seedling. The hairs increase the surface area for absorption of water.

KEY POINT

If a plant needs a mineral ion that is at a low concentration in the soil, the root hair cells use active transport to absorb it.

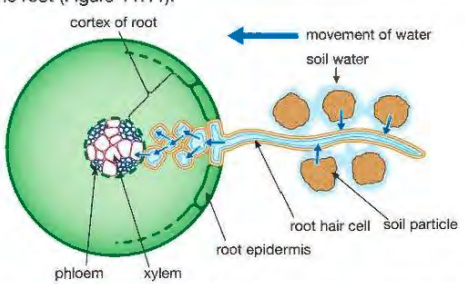
BIOLOGY ONLY

UPTAKE OF WATER BY ROOTS

The regions just behind the growing tips of the roots of a plant are covered in thousands of tiny **root hairs** (Figure 11.10). These areas are the main sites of water absorption by the roots, where the hairs greatly increase the surface area of the root epidermis.

Each hair is actually a single, specialised cell of the root epidermis. The long, thin outer projection of the root hair cell penetrates between the soil particles, reaching the soil water. The water in the soil has some solutes dissolved in it, such as mineral ions, but their concentrations are much lower than the concentrations of solutes inside the root hair cell.

The soil water therefore has a higher water potential than the inside of the cell. This allows water to enter the root hair cell by osmosis. In turn, this water movement dilutes the contents of the cell, increasing its water potential. Water then moves out of the root hair cell into the outer tissue of the root (called the root cortex). Continuing in this way, a gradient of water potential is set up across the root cortex, kept going by water being taken up by the xylem in the middle of the root (Figure 11.11).



▲ Figure 11.11 Water is taken up by root hairs of the plant epidermis and carried across the root cortex by a water potential gradient. It then enters the xylem and is transported to all parts of the plant.

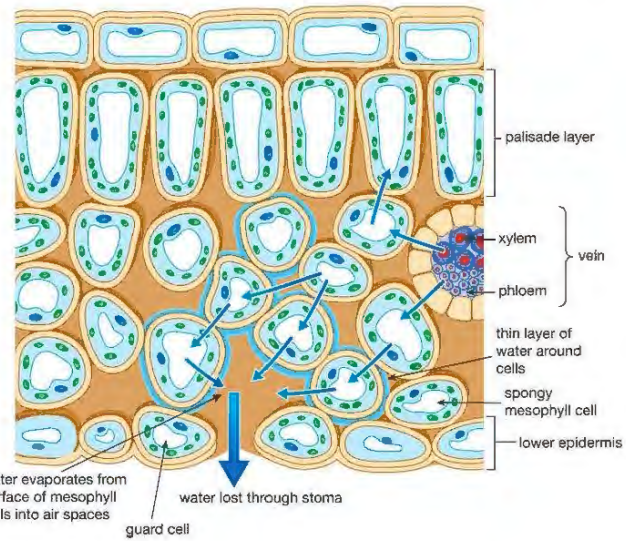
DID YOU KNOW?

Plants that live in dry habitats, such as cacti and succulents (plants with fleshy leaves), usually have a very thick layer of waxy cuticle on their leaves. Cacti have leaves reduced to spines. which have a low surface area. These adaptations reduce water loss.

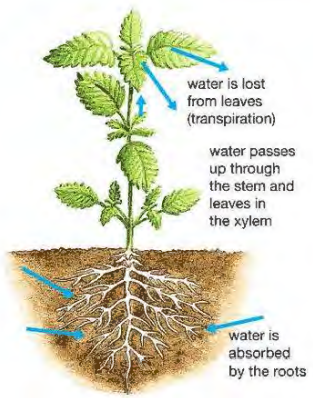
LOSS OF WATER BY THE LEAVES – TRANSPIRATION

Osmosis is also involved in the movement of water through leaves. The epidermis of leaves is covered by a waxy cuticle (see Chapter 10), which is impermeable to water. Most water passes out of the leaves as water vapour through pores called stomata (stoma is the singular of stomata).

Water leaves the cells of the leaf mesophyll and evaporates into the air spaces between the spongy mesophyll cells. The water vapour then diffuses out through the stomatal pores (Figure 11.12).



▲ Figure 11.12 Passage of water from the xylem to the stomatal pores of a leaf



▲ Figure 11.13 The transpiration stream



▲ Figure 11.14 Xylem vessels in a stem. The spirals and rings (stained red) are made of lignin.

Loss of water from the mesophyll cells sets up a water potential gradient which 'draws' water by osmosis from surrounding mesophyll cells. In turn, the xylem vessels supply the leaf mesophyll tissues with water.

This loss of water vapour from the leaves is called **transpiration**. Transpiration causes water to be 'pulled up' the xylem in the stem and roots in a continuous flow known as the **transpiration stream** (Figure 11.13). The transpiration stream has more than one function. It:

- supplies water for the leaf cells to carry out photosynthesis
- carries mineral ions dissolved in the water
- provides water to keep the plant cells turgid
- allows evaporation from the leaf surface, which cools the leaf, in a similar way to sweat cooling the human skin.

END OF BIOLOGY ONLY

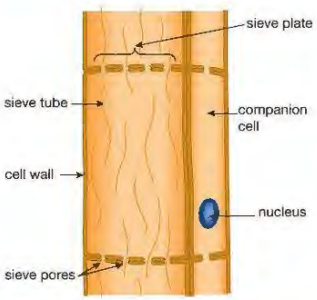
TRANSPORT IN THE XYLEM

Xylem transports water and minerals throughout the plant. Xylem contains dead cells arranged end-to-end, forming continuous vessels. When they are mature, the vessels contain no cytoplasm. Instead, they have a hollow central space or **lumen** through which the water passes. The walls of the xylem vessels contain a woody material called **lignin** (Figure 11.14).

The xylem vessels begin life as living cells with normal cytoplasm and cellulose cell walls. As they develop they become elongated, and gradually their original cellulose cell walls become impregnated with lignin, made by the cytoplasm. As this happens, the cells die, forming hollow tubes. Lignification makes them very strong, and enables them to carry water up tall plants without collapsing. Lignin is also impermeable to water.

TRANSPORT IN THE PHLOEM

The other plant transport tissue, the phloem, consists of living cells at all stages in its development. Tubes in the phloem are also formed by cells arranged end-to-end, but they have cell walls made of cellulose, and retain their cytoplasm. The end of each cell is formed by a cross-wall of cellulose with holes, called a **sieve plate**. The living cytoplasm extends through the holes in the sieve plates, linking each cell with the next, forming a long **sieve tube** (Figure 11.15). The sieve tubes transport the products of photosynthesis from the leaves to other parts of the plant. Sugars for energy, or amino acids for building proteins, are carried to young leaves and other growing points in the plant. Sugar may also be taken to the roots and converted into starch for storage. Despite being living cells, the phloem sieve tubes have no nucleus. They seem to be controlled by other cells that lie alongside the sieve tubes, called **companion cells** (Figure 11.15).



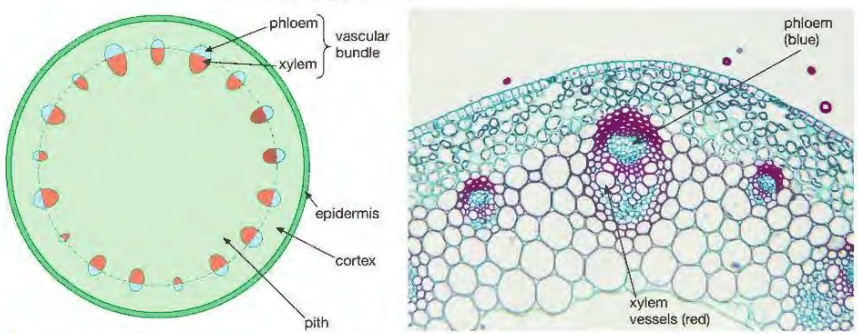
▲ Figure 11.15 Xylem vessels in a stem

REMINDER

Remember: xylem carries water and minerals up from the roots. Phloem carries products of photosynthesis away from the leaves. The contents of the phloem can travel up or down the plant.

STRUCTURE OF A STEM

In a young stem, xylem and phloem are grouped together in areas called **vascular bundles**. Unlike in the root, where the vascular tissue is in the central core, the vascular bundles are arranged in a circle around the outer part of the stem (Figure 11.16).



▲ Figure 11.16 (a) This cross-section of a stem shows the arrangement of xylem and phloem tissue in vascular bundles. (b) Three vascular bundles in part of a stem section. The outer red cells are lignified fibres for extra support.

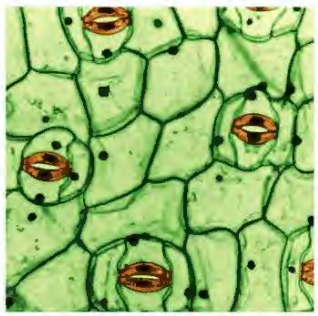
KEY POINT

'Vascular' means 'made of vessels'. A vascular bundle is a group of vessels or tubes (xylem and phloem).

In older stems, the vascular tissue grows to form complete rings around the stem. The inner xylem forms the woody central core of a stem, with the living layer of phloem outside this.

BIOLOGY ONLY

CONTROL OF TRANSPIRATION BY STOMATA



▲ Figure 11.17 Stomata in the surface of the lower epidermis of a leaf

DID YOU KNOW?

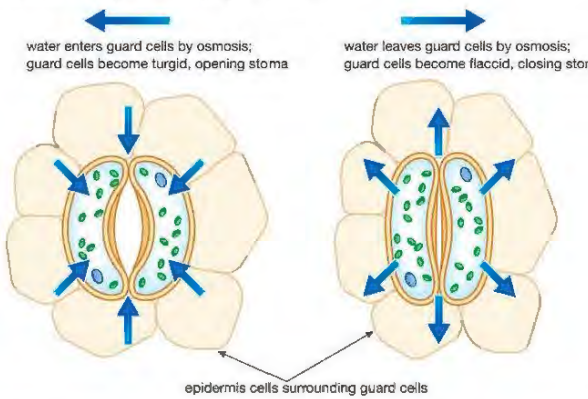
You can model the action of a guard cell using a long balloon and some sticky tape. Stick a few pieces of tape down one side of the balloon and then blow it up. As it inflates, the balloon will curve outwards like a turgid guard cell. The tape represents the thick inner cell wall.

EXTENSION WORK

You can make a 'mould' of a leaf surface using colourless nail varnish. Paint an area of nail varnish about 1 cm^2 on the lower surface of a leaf. Leaves that are smooth, hair-free, and not too delicate work well, such as those of privet (*Ligustrum sp.*). Leave the nail varnish to dry for half an hour and then use forceps to peel it off the leaf. Carefully place the thin film of varnish flat on a slide and look at it through a microscope (you don't need to use a cover slip). Under medium or high power you should be able to see impressions of the stomata and guard cells.

As you saw in Chapter 10, there are usually more stomata on the lower surface of the leaves than the upper surface in most plant species (Figure 11.17). If they were mainly on the upper leaf surface, the leaf would lose too much water. This is because the stomata would be exposed to direct sunlight, which would produce a high rate of evaporation from them. There is also less air movement on the underside of leaves. The evolution of this arrangement of stomata is an adaptation that reduces water loss.

The stomata can open and close. The guard cells that surround each stoma have an unusual 'banana' shape, and the part of their cell wall nearest the stoma is especially thick. In the light, water enters the guard cells by osmosis from the surrounding epidermis cells. This causes the guard cells to become turgid, and, as they swell up, their shape changes. They bend outwards, opening up the stoma. In the dark, the guard cells lose water again, they become flaccid and the stoma closes. No one knows for sure how this change is brought about, but it seems to be linked to the fact that the guard cells are the only cells in the lower epidermis that contain chloroplasts. In the light, the guard cells use energy to accumulate solutes in their vacuoles, causing water to be drawn in by osmosis (Figure 11.18).



▲ Figure 11.18 When the guard cells become turgid, the stoma opens. When they become flaccid, it closes.

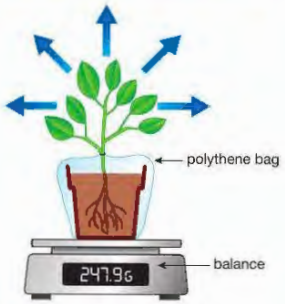
Closure of stomata in the dark is a useful adaptation. Without the sun there is no need for loss of water vapour from the stomata to cool the leaves. In addition, leaves cannot photosynthesise in the dark, so they don't need water for this purpose. Therefore, it doesn't matter if the transpiration stream is shut down by closure of the stomata. As you will see, other physical factors can also affect the rate of transpiration.

FACTORS AFFECTING THE RATE OF TRANSPIRATION

There are four main factors which affect the rate of transpiration:

- **Light intensity** The rate of transpiration increases in the light, because of the opening of the stomata in the leaves, so that the leaf can photosynthesise.
- **Temperature** High temperatures increase the rate of transpiration, by increasing the rate of evaporation of water from the mesophyll cells.
- **Humidity** When the air around the plant is humid, this reduces the diffusion gradient between the air spaces in the leaf and the external air. The rate of transpiration therefore decreases in humid air and speeds up in dry air.
- **Wind speed** The rate of transpiration increases with faster air movements across the surface of the leaf. The moving air removes any water vapour which might remain near the stomata. This moist air would otherwise reduce the diffusion gradient and slow down diffusion.

MEASURING THE RATE OF TRANSPIRATION: POTOMETERS



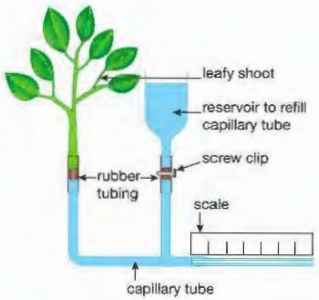
▲ Figure 11.19 A 'weight' potometer

A potometer is a simple piece of apparatus which measures the rate of transpiration or the rate of uptake of water by a plant. (These are not the same thing – some of the water taken up by the plant may stay in the plant cells, or be used for photosynthesis.) There are two types, 'weight' and volume potometers.

A 'weight' potometer measures the rate of loss of mass from a potted plant or leafy shoot over an extended period of time, usually several hours (Figure 11.19).

The polythene bag around the pot prevents loss of moisture by evaporation from the soil. Most of the mass lost by the plant will be due to water evaporating from the leaves during transpiration. (However, there will be small changes in mass due to respiration and photosynthesis, since both of these processes exchange gases with the air).

A volume potometer is used to find the rate of uptake of water by a leafy shoot, by 'magnifying' this uptake in a capillary tube. These potometers come in various shapes and sizes. The simplest is a straight vertical tube joined to the shoot by a piece of rubber tubing. More complex versions have a horizontal capillary tube and a way of refilling the capillary to re-set the water at its starting position (Figure 11.20).



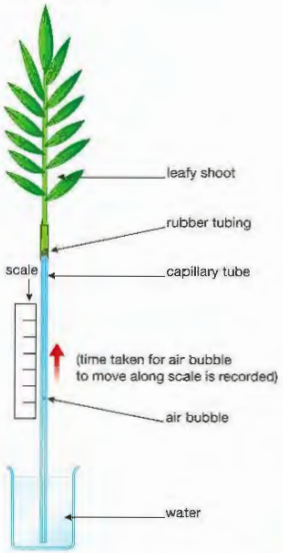
▲ Figure 11.20 A commercial potometer with a water reservoir

> ! Safety Note: Wash hands after assembling the plant stem. Do not handle a hair dryer, fan or light with wet hands.

ACTIVITY 3

▼ PRACTICAL: INVESTIGATING THE EFFECTS OF ENVIRONMENTAL FACTORS ON THE RATE OF TRANSPIRATION USING A SIMPLE POTOMETER

The capillary and attached rubber tubing is placed in a sink of water and any air it contains removed by squeezing the tubing. A shoot is taken from a plant and the end of the shoot is cut at an angle (this is done under water to stop air entering the shoot). The angled cut makes it easier to push the stem into the rubber tubing. The assembled apparatus is allowed to dry and any joints sealed with petroleum jelly. The simple potometer is clamped in a vertical position and left to adjust to its new conditions (Figure 11.21).



◀ Figure 11.21 A simple potometer

The distance moved by the column of water in the capillary tube is measured over a certain time, e.g. 10 minutes, and the rate of movement calculated. The plant is then exposed to different environmental conditions to see how they affect the rate. For example:

- Using a fan or hair dryer to investigate the effect of moving air
- Placing the plant under a bright light to investigate the effect of changing light intensity.

For a controlled experiment, only one condition is changed at a time. Each time a condition is changed, the shoot is left to adjust or equilibrate to the new conditions. How can you tell when the shoot has equilibrated to the change? (Hint – what will happen to the rate of transpiration?)

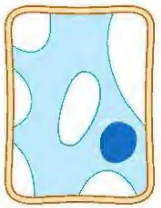
END OF BIOLOGY ONLY

CHAPTER QUESTIONS

More questions on transport in plants can be found at the end of Unit 3 on page 182.

SKILLS CRITICAL THINKING (8)

1 A plant cell was placed in a concentrated sucrose solution. The diagram below shows the appearance of the cell after an hour in the solution.



What is the best description of this cell?

A turgid
B flaccid
C plasmolysed
D wilted

(4)

2 What are the functions of xylem and phloem?

A xylem transports water and sugars, phloem transports mineral ions
B xylem transports water and mineral ions, phloem transports sugars
C xylem transports sugars and mineral ions, phloem transports water
D xylem transports sugars, phloem transports water and mineral ions

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SKILLS CRITICAL THINKING (4)

3 In the table below, which row shows the correct directions of movement of materials in the xylem and phloem?

	Xylem	Phloem
A	up and down	down only
B	down only	up and down
C	up only	up and down
D	up and down	up only

(6)

4 Which of the following conditions result in the highest rate of transpiration?

A warm, windy and dry
B warm, windy and wet
C cold, windy and dry
D warm, still and wet

END OF BIOLOGY ONLY

SKILLS PROBLEM SOLVING (6)

5 Three 'chips' of about the same size and shape were cut from the same potato. Each was blotted, weighed and placed in a different sucrose solution (A, B or C). The chips were left in the solutions for an hour, then removed, blotted and re-weighed. Here are the results:

	Starting mass / g	Final mass / g	Change in mass / %
solution A	7.4	6.5	−12.2
solution B	8.2	8.0	
solution C	7.7	8.5	+10.4

a Calculate the percentage change in mass for the chip in solution B.

SKILLS CRITICAL THINKING (7)

b Name the process that caused the chips to lose or gain mass.

SKILLS ANALYSIS (8)

c Which solution was likely to have been the most concentrated?
d Which solution had the highest water potential?
e Which solution had a water potential most similar to the water potential of the potato cells?

SKILLS CRITICAL THINKING (7)

f The cell membrane is described as 'partially permeable'. Explain the meaning of this.

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SKILLS CRITICAL THINKING (6)

6 Explain how each of the following cells is adapted for its function:
a a root hair cell b a xylem vessel c a guard cell.

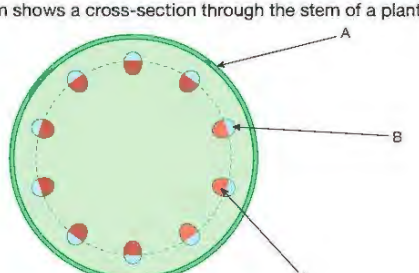
SKILLS REASONING (7)

7 Suggest reasons for each of the following observations.
a When removing a small plant from a pot and re-planting it in a garden, it is important to take the plant out of its pot carefully, leaving the roots in a ball of soil. If this is not done, the plant may wilt after it has been transplanted.
b A plant cutting is more likely to grow successfully if you remove some of its leaves before planting it in compost.
c Plants that live in very dry habitats often have stomata located in sunken pits in their leaves.
d Greenflies feed by sticking their hollow tube-like mouthparts into the phloem of a plant stem.

END OF BIOLOGY ONLY

SKILLS ANALYSIS (5)

8 The diagram shows a cross-section through the stem of a plant.



a Identify the tissues labelled A, B and C.

SKILLS REASONING 6

b A young stem was placed in a solution of a red dye for an hour. Which tissue in the diagram would be most likely to contain the dye? Explain your answer.

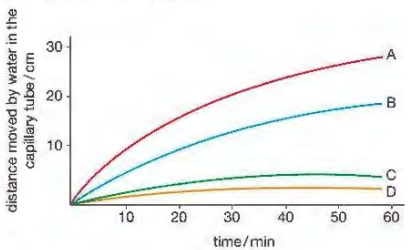
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SKILLS ANALYSIS, INTERPRETATION 7

9 A simple volume potometer was used to measure the uptake of water by a leafy shoot under four different conditions. During the experiment, the temperature, humidity and light intensity were kept constant. The conditions were:

1. Leaves in still air with no petroleum jelly applied to them.
2. Leaves in moving air with no petroleum jelly applied to them.
3. Leaves in still air with the lower leaf surface covered in petroleum jelly.
4. Leaves in moving air with the lower surface covered in petroleum jelly.

The results are shown in the graph.



a Copy and complete the table below to show which condition (1, 2, 3 or 4) is most likely to produce curve A, B, C and D. One has been done for you:

Condition	Curve
1	B
2	
3	
4	

SKILLS REASONING 6

b Explain why moving air affects the rate of uptake of water by the shoot.

SKILLS CRITICAL THINKING 5

10 During transpiration, water moves through the cells of a leaf and is finally lost through the stomata by evaporation.

a Explain how water travels from one cell to another in the leaf.

SKILLS REASONING 6

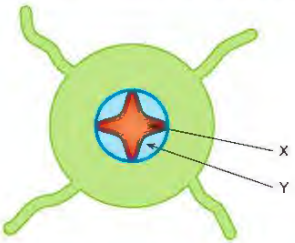
b How would the rate of transpiration be affected if the air temperature increased from 20 °C to 30 °C? Explain your answer.

SKILLS CRITICAL THINKING 6

c Describe an adaptation found in plants living in dry habitats that decreases water loss from the leaves.

SKILLS INTERPRETATION 5

11 The diagram shows a section through the root of a plant.



a Identify the tissues labelled X and Y.

6

b Draw and label a root hair cell.

SKILLS CRITICAL THINKING 8

c Describe how a root hair cell absorbs water.

SKILLS INTERPRETATION 8

12 Write a short description of how water moves from the soil through the plant. Your description must include these words: xylem, evaporation, root hair cells, water potential, stomata and osmosis.

Underline each of these words in your description.

END OF BIOLOGY ONLY